

Conic Conformal Geometric Algebra: Foundation for Two-Dimensional Conic Geometry

Enzo Harquin, Stéphane Breuils, Venceslas Biri and Vincent Nozick

Abstract. We present Conic Conformal Geometric Algebra (CCGA), the geometric algebra $\mathbb{R}_{5,3}$ for the geometry of planar conics, as a systematic foundation for two-dimensional conic geometry. CCGA renames and consolidates the algebra previously introduced as QC2GA and proven isomorphic to the Geometric Algebra for Conics (GAC). *[TODO: finalise abstract once sections are drafted.]* We give the complete catalogue of geometric objects in both outer-product null space (OPNS) and inner-product null space (IPNS) form, with their grades and reality conditions; the join, meet, and dual operations; closed-form, geometric-algebra-native extraction of every classical conic feature directly from the dual conic vector; the versor transformation group together with a precise account of which planar maps it realises; and a unified treatment of conic intersection based on a shared-factor reduction of the regressive product. Every construction is given in closed geometric-algebra form with its reality conditions.

Keywords. Conics, Geometric Algebra, Clifford Algebra, Conic Conformal Geometric Algebra (CCGA), Geometric Algebra for Conics (GAC), Projective Geometry, OPNS/IPNS duality, Conic intersection.

1. Introduction

Geometric Algebra (GA) provides ... *[TODO: write 2–3 paragraphs of motivation, adapting the introduction of the base paper to a foundations framing.]*

1.1. Contributions

The main contributions of this paper are:

- A self-contained construction of **Conic Conformal Geometric Algebra** (CCGA, $\mathbb{R}_{5,3}$), unifying the former QC2GA and GAC under one name and one notation (Sections 2 and 3).
- The **complete object taxonomy** of CCGA in both OPNS and IPNS form, with grades, geometric meaning, and reality conditions, organised by a single construction ladder (Sections 4 and 5).
- **Geometric-algebra-native feature extraction**: closed-form formulas reading the type, center, axes, foci, eccentricity, asymptotic directions, and tangent/polar lines of a conic directly from its dual grade-1 vector by inner products (Section 7).
- A precise characterisation of the **versor transformation group** of CCGA — which planar maps are sandwich versors (similarities) and which are only linear outermorphisms (shears, non-uniform scalings, projective maps) (Section 8).
- A unified **intersection framework** based on a shared-factor / Bézout reduction of the regressive product, recovering and organising the pencil method of the base paper (Section 9).

1.2. Notation and Conventions

We follow the GA conventions of [3, 5]. *[TODO: notation table.]* Throughout, $A \equiv B$ means A equals B up to a nonzero scalar, $\langle A \rangle_k$ is the grade- k part, and the dual is fixed as $A^* = A I^{-1}$ (right multiplication by the inverse pseudoscalar). Lower-case bold letters denote basis blades and multivectors ($\mathbf{a}$); italic lower-case letters denote their scalar components ($a_1, x, y^2, \dots$); the superscript star $\mathbf{x}^*$ is the dual; and the subscript ϵ in $\mathbf{x}_\epsilon$ extracts the Euclidean (e_1, e_2) part of a vector. The contraction and outer products take priority over the full geometric product, e.g. $\mathbf{a} \wedge \mathbf{b} \mathbf{I} = (\mathbf{a} \wedge \mathbf{b}) \mathbf{I}$.

2. The CCGA Algebra $\mathbb{R}_{5,3}$

CCGA is the geometric algebra $\mathbb{R}_{5,3}$ (five positive, three negative, no null basis axes), of dimension $2^8 = 256$. It is the conic algebra first introduced by Perwass [5], developed as the two-dimensional restriction QC2GA of Quadric Conformal Geometric Algebra [1] and shown isomorphic to the Geometric Algebra for Conics (GAC) [4]; we call it Conic Conformal Geometric Algebra and relate the two bases in Section 2.6. Because the metric is *non-degenerate*, the pseudoscalar is invertible, so the dual, meet, and join are all clean — in contrast with the degenerate metric of PGA, where these require special care.

2.1. Diagonal and null bases

The orthogonal basis splits into the Euclidean pair $\{e_1, e_2\}$ (+1), three positive vectors $\{e_{+1}, e_{+2}, e_{+3}\}$ (+1), and three negative vectors $\{e_{-1}, e_{-2}, e_{-3}\}$ (−1). The null working basis is, for $1 \leq i \leq 3$,

$$e_{o_i} = e_{+i} + e_{-i}, \quad e_{\infty_i} = \frac{1}{2} (e_{-i} - e_{+i}), \quad (1)$$

with inverse $e_{+i} = \frac{1}{2}e_{o_i} - e_{\infty_i}$, $e_{-i} = \frac{1}{2}e_{o_i} + e_{\infty_i}$. Each pair (e_{o_i}, e_{∞_i}) is null, $e_{o_i}^2 = e_{\infty_i}^2 = 0$, with hyperbolic pairing $e_{o_i} \cdot e_{\infty_i} = -1$, and the three pairs are mutually orthogonal. The resulting Gram matrix is

$$\begin{array}{c|cccccccc}
 & e_1 & e_2 & e_{o_1} & e_{\infty_1} & e_{o_2} & e_{\infty_2} & e_{o_3} & e_{\infty_3} \\
\hline
e_1 & 1 & . & . & . & . & . & . & . \\
e_2 & . & 1 & . & . & . & . & . & . \\
e_{o_1} & . & . & 0 & -1 & . & . & . & . \\
e_{\infty_1} & . & . & -1 & 0 & . & . & . & . \\
e_{o_2} & . & . & . & . & 0 & -1 & . & . \\
e_{\infty_2} & . & . & . & . & -1 & 0 & . & . \\
e_{o_3} & . & . & . & . & . & . & 0 & -1 \\
e_{\infty_3} & . & . & . & . & . & . & -1 & 0
\end{array} \tag{2}$$

This Gram matrix follows directly from the null-basis definitions (1).

Remark 2.1. A symmetric normalisation $e_{o_i} = \frac{1}{\sqrt{2}}(e_{-i} - e_{+i})$, $e_{\infty_i} = \frac{1}{\sqrt{2}}(e_{+i} + e_{-i})$ yields the same metric of Equation (2) and is convenient for deriving rotation/translation/scaling versors. We adopt the asymmetric convention (1) throughout, as it makes the point embedding (8) carry the conic coefficients directly; the two differ only by a fixed rescaling of each null pair.

2.2. Special blades and the pseudoscalar

The following blades structure every later construction:

$$e_o = e_{o_1} + e_{o_2}, \quad e_\infty = \frac{1}{2}(e_{\infty_1} + e_{\infty_2}), \tag{3a}$$

$$\bar{e}_o = e_{o_1} - e_{o_2}, \quad \bar{e}_\infty = \frac{1}{2}(e_{\infty_1} - e_{\infty_2}), \tag{3b}$$

$$I_o^\triangleright = \bar{e}_o \wedge e_{o_3}, \quad I_\infty^\triangleright = (e_{\infty_1} - e_{\infty_2}) \wedge e_{\infty_3}, \tag{3c}$$

$$I_o = e_{o_1} \wedge e_{o_2} \wedge e_{o_3}, \quad I_\infty = e_{\infty_1} \wedge e_{\infty_2} \wedge e_{\infty_3}, \tag{3d}$$

$$I_\epsilon = e_1 \wedge e_2, \quad I = I_\epsilon \wedge I_\infty \wedge I_o. \tag{3e}$$

Table 1 summarises these blades, their grades, and their geometric role.

TABLE 1. The special blades of CCGA, with grades and geometric role (Section 2.3).

Symbol	Definition	Grade	Role
e_o	$e_{o_1} + e_{o_2}$	1	combined origin
e_∞	$\frac{1}{2}(e_{\infty_1} + e_{\infty_2})$	1	isotropic infinity
$\bar{e}_o$	$e_{o_1} - e_{o_2}$	1	differential origin
$\bar{e}_\infty$	$\frac{1}{2}(e_{\infty_1} - e_{\infty_2})$	1	differential infinity
$I_o^\triangleright$	$\bar{e}_o \wedge e_{o_3}$	2	origin gauge
$I_\infty^\triangleright$	$(e_{\infty_1} - e_{\infty_2}) \wedge e_{\infty_3}$	2	infinity gauge (circular points)
I_o	$e_{o_1} \wedge e_{o_2} \wedge e_{o_3}$	3	origin 3-blade
I_∞	$e_{\infty_1} \wedge e_{\infty_2} \wedge e_{\infty_3}$	3	line/conic at infinity
I_ϵ	$e_1 \wedge e_2$	2	Euclidean pseudoscalar
I	$I_\epsilon \wedge I_\infty \wedge I_o$	8	pseudoscalar

Proposition 2.2. *The grade-1 special blades satisfy*

$$e_o \cdot e_\infty = -1, \quad \bar{e}_o \cdot \bar{e}_\infty = -1, \quad e_{o_3} \cdot e_{\infty_3} = -1, \quad (4)$$

and the two pairs are mutually orthogonal, $\{e_o, e_\infty\} \perp \{\bar{e}_o, \bar{e}_\infty\}$. The pseudoscalar squares to $I^2 = -1$, hence $I^{-1} = -I$ and $II^{-1} = 1$.

These relations follow by direct computation from Equation (2).

2.3. Geometric meaning of the special blades

The eight null directions group into the three conjugate pairs of Equation (2), each carrying a distinct piece of conic structure. On the *origin* side, $e_o = e_{o_1} + e_{o_2}$ is the combined origin, $\bar{e}_o = e_{o_1} - e_{o_2}$ the *differential* origin, and e_{o_3} the *cross-term* origin; the *infinity* side mirrors them with the isotropic e_∞ , the differential $\bar{e}_\infty$, and the cross-term e_{∞_3} . The three pairs govern, respectively, the *size*/isotropy of a round object (the radius, carried by e_∞), the $x^2 - y^2$ *anisotropy* that distinguishes the two axes (carried by $\bar{e}_o, \bar{e}_\infty$), and the *xy tilt* (carried by e_{o_3}, e_{∞_3}).

This split is the algebraic form of the principle that *shape lives on the origin side and size on the infinity side*: in the dual conic vector (13), the quadratic coefficients A, B, C sit on $e_{o_1}, e_{o_2}, e_{o_3}$ while the constant F sits on $e_{\infty_1} + e_{\infty_2}$. The infinity-gauge blade $I_\infty^\flat = (e_{\infty_1} - e_{\infty_2}) \wedge e_{\infty_3}$ is the pair of *circular points* I, J at infinity through which every circle passes; it controls the round sub-family and the intersection reductions of Section 9.

2.4. Subspace structure

The eight-dimensional space splits as $\mathbb{R}^8 = V_6 \oplus W_2$, where

$$V_6 = \text{span}\{e_o, e_1, e_2, e_{\infty_1}, e_{\infty_2}, e_{\infty_3}\}, \quad W_2 = \text{span}\{\bar{e}_o, e_{o_3}\}. \quad (5)$$

V_6 is the *point/conic space* — it carries the six Veronese coordinates $(1, x, y, x^2, y^2, xy)$ of a conic — while W_2 is the two-dimensional *gauge plane*, whose top blade is exactly $I_o^\flat = \bar{e}_o \wedge e_{o_3}$. This is why $I_o^\flat$ is the gauge blade used to fix the outer-product conic to grade 7 (Section 4): it supplies precisely the two directions that the points cannot reach. The proof that the point embedding (8) lands in V_6 (and never in W_2) is given in Section 3.

2.5. Reciprocity identities

The two gauge blades $I_o^\flat$ and $I_\infty^\flat$ are reciprocal, and the line at infinity factors through $I_\infty^\flat$:

$$I_\infty^\flat \wedge e_\infty = -I_\infty, \quad e_{\infty_3} \wedge \bar{e}_\infty = -\frac{1}{2} I_\infty^\flat, \quad I_\infty^\flat \lrcorner I_o^\flat = -2. \quad (6)$$

The last identity makes the contraction $\lrcorner I_\infty^\flat$ the inverse of the wedge $\wedge I_o^\flat$, which is the algebraic mechanism behind the conic-intersection pipeline of Section 9; the first lets a line, as a degenerate conic, carry the whole line at infinity I_∞ .

2.6. Relation to GAC

CCGA and the Geometric Algebra for Conics (GAC) [4] are the same algebra in different bases. Writing the GAC basis $(e_1, e_2, \bar{n}_+, n_+, \bar{n}_-, n_-, \bar{n}_\times, n_\times)$ in terms of the CCGA null basis,

$$\bar{n}_+ = e_o, \quad n_+ = e_\infty, \quad \bar{n}_- = \bar{e}_o, \quad n_- = \bar{e}_\infty, \quad \bar{n}_\times = e_{o_3}, \quad n_\times = e_{\infty_3}, \quad (7)$$

exhibits the isomorphism as a pure change of basis; the full equivalence is established in [2]. The GAC names match the geometric roles of Section 2.3: the $+$ pair is isotropic (size), the $-$ pair is the $x^2 - y^2$ anisotropy, and the $\times$ pair is the xy cross term.

3. The Point Embedding

The conic point map $\mathbb{R}^2 \rightarrow \mathbb{R}^8$ is the Veronese-type embedding

$$\mathbf{p} = e_o + \mathbf{p}_\epsilon + \frac{1}{2} (x^2 e_{\infty_1} + y^2 e_{\infty_2}) + xy e_{\infty_3}, \quad \mathbf{p}_\epsilon = x e_1 + y e_2. \quad (8)$$

Note that the null vector e_{o_3} does not appear in a point: this preserves the C2GA property that the inner product of two points is (minus half) their squared Euclidean distance. For $\mathbf{p}_1, \mathbf{p}_2$ at $(x_1, y_1), (x_2, y_2)$,

$$\begin{aligned} \mathbf{p}_1 \cdot \mathbf{p}_2 &= -\frac{x_1^2}{2} - \frac{x_2^2}{2} - \frac{y_1^2}{2} - \frac{y_2^2}{2} + x_1 x_2 + y_1 y_2 \\ &= -\frac{1}{2} ((x_1 - x_2)^2 + (y_1 - y_2)^2) = -\frac{1}{2} \text{Dist}^2(\mathbf{p}_1, \mathbf{p}_2), \end{aligned} \quad (9)$$

so in particular $\mathbf{p}^2 = 0$ (every point is null).

4. Conics: OPNS and IPNS

4.1. The grade-1 IPNS conic

$$\mathbf{p} \cdot \mathbf{s} = 0 \iff Ax^2 + By^2 + Cxy + Dx + Ey + F = 0. \quad (10)$$

4.2. The grade-7 OPNS conic and the $I_o^\flat$ gauge

$$\mathbf{c} = I_o^\flat \wedge \mathbf{p}_1 \wedge \cdots \wedge \mathbf{p}_5, \quad \mathbf{c}^* = \mathbf{c} I^{-1} \text{ is the grade-1 conic vector.} \quad (11)$$

Expanding the five-point wedge and collecting the seven blade components — two of which share a coefficient — gives the closed blade form of $\mathbf{c}$ whose dual is the grade-1 conic vector $\mathbf{c}^*$ of Equation (10):

$$\begin{aligned} \mathbf{c} &= 2A(I_\epsilon \wedge e_{\infty_2} \wedge e_{\infty_3} \wedge I_o) + 2B(I_\epsilon \wedge e_{\infty_1} \wedge e_{\infty_3} \wedge I_o) + C(I_\epsilon \wedge e_{\infty_1} \wedge e_{\infty_2} \wedge I_o) \\ &\quad - D(e_2 \wedge I_\infty \wedge I_o) - E(e_1 \wedge I_\infty \wedge I_o) - \frac{1}{2}F(I_\epsilon \wedge I_\infty \wedge e_o) = \mathbf{c}^* I, \end{aligned} \quad (12)$$

$$\mathbf{c}^* = -(2A e_{o_1} + 2B e_{o_2} + C e_{o_3}) + D e_1 + E e_2 - F e_\infty. \quad (13)$$

The reading $\mathbf{p} \cdot \mathbf{c}^* = 0 \iff \mathbf{p} \wedge \mathbf{c} = 0$ recovers the locus (10), so the six conic coefficients sit directly in the grade-1 vector (13).

4.3. Axis-aligned conics by infinity gauge

An axis-aligned conic $Ax^2 + By^2 + Dx + Ey + F = 0$ has no cross term, i.e. $C = 0$, leaving five coefficients and four degrees of freedom; it can therefore be built from *four* points. The cross term is removed by wedging with e_{∞_3} , which annihilates the xy carrier of every point:

$$p \wedge e_{\infty_3} = (e_o + p_\epsilon + \frac{1}{2}(x^2 e_{\infty_1} + y^2 e_{\infty_2})) \wedge e_{\infty_3}. \quad (14)$$

Hence an axis-aligned conic is a general conic with four points sent to the e_{∞_3} direction:

$$\mathbf{c} = I_o^\flat \wedge p_1 \wedge p_2 \wedge p_3 \wedge p_4 \wedge e_{\infty_3}. \quad (15)$$

Equivalently, $C = 0$ is the statement that the e_{o_3} coefficient of $\mathbf{c}^*$ in (13) vanishes.

The infinity gauge. The four points in (15) carry one redundant degree of freedom: the *type* of a conic is decided entirely by its two points at infinity — its intersection with the line at infinity I_∞ . Spending that degree of freedom, by replacing the fourth point with a fixed grade-2 *infinity gauge* blade B_∞ , produces an axis-aligned conic of prescribed type from only three points:

$$\mathbf{c} = p_1 \wedge p_2 \wedge p_3 \wedge B_\infty \wedge I_o^\flat, \quad B_\infty = (\alpha e_{\infty_1} + \beta e_{\infty_2}) \wedge e_{\infty_3}. \quad (16)$$

B_∞ is the pair of ideal points of $\mathbf{c}$, the two solutions of its quadratic part on I_∞ ; choosing $B_\infty = p_4 \wedge e_{\infty_3}$ recovers the general axis-aligned conic (15). The smooth types are listed in Table 2.

TABLE 2. Axis-aligned conics by infinity gauge B_∞ (smooth types).

Type	B_∞	ideal points	Δ_2
circle	$(e_{\infty_1} - e_{\infty_2}) \wedge e_{\infty_3} = I_\infty^\flat$	imaginary $(1, \pm i)$	$> 0, A=B$
ellipse $a:b$	$(a^2 e_{\infty_1} - b^2 e_{\infty_2}) \wedge e_{\infty_3}$	imaginary $(a, \pm ib)$	> 0
hyperbola $a:b$	$(a^2 e_{\infty_1} + b^2 e_{\infty_2}) \wedge e_{\infty_3}$	real $(a, \pm b)$	< 0
parabola $\parallel x$	$e_{\infty_1} \wedge e_{\infty_3}$	real double $(1, 0)$	0
parabola $\parallel y$	$e_{\infty_2} \wedge e_{\infty_3}$	real double $(0, 1)$	0

Proposition 4.1 (type from the infinity gauge). *The conic (16) has $C = 0$ and quadratic coefficients $A \propto \alpha$, $B \propto -\beta$, hence discriminant $\Delta_2 = AB \propto -\alpha\beta$. Therefore*

$$\alpha\beta < 0 \Rightarrow \Delta_2 > 0 \text{ (ellipse)}, \quad \alpha\beta = 0 \Rightarrow \Delta_2 = 0 \text{ (parabola)}, \quad \alpha\beta > 0 \Rightarrow \Delta_2 < 0 \text{ (hyperbola)} \quad (17)$$

which is the projective classification by incidence with I_∞ : an imaginary, double, or real ideal pair (the discriminants Δ_2, Δ_3 are defined in Section 7).

The ideal pairs are wedges of the at-infinity points of Section 3: the real pair $v_\infty(a, b) \wedge v_\infty(a, -b) \equiv (a^2 e_{\infty_1} + b^2 e_{\infty_2}) \wedge e_{\infty_3}$ (hyperbola), its imaginary counterpart $(a^2 e_{\infty_1} - b^2 e_{\infty_2}) \wedge e_{\infty_3}$ (ellipse), and the double contact $v_\infty(1, 0) \wedge v'_\infty(1, 0) \equiv e_{\infty_1} \wedge e_{\infty_3}$ (parabola). The circle is the isotropic case

$\alpha = -\beta$, where $B_\infty = I_\infty^\perp$ is the circular-point pair of Section 2.3; the construction then specialises to the circle, and (16) to the standard axis-aligned ellipse, up to scale.

Degenerate conics. Within each Δ_2 class a degenerate member ($\Delta_3 = 0$) appears when the three points are placed in special position; B_∞ still fixes Δ_2 , while Δ_3 drops to zero:

- *Two crossing lines* (degenerate hyperbola): $B_\infty = (e_{\infty_1} + e_{\infty_2}) \wedge e_{\infty_3}$ with the three points through the common centre, e.g. yielding $x^2 - y^2 = 0$.
- *Two parallel lines* (degenerate parabola): $B_\infty = e_{\infty_2} \wedge e_{\infty_3}$ with three points on two vertical lines, e.g. $x^2 - 1 = 0$.
- *A point* (degenerate ellipse, two imaginary lines): the $r \rightarrow 0$ limit of an imaginary circle, $x^2 + y^2 = 0$; having no three distinct real points, it arises only as a limit.

Since the type test keys on Δ_2 alone, a degenerate hyperbola still reports as a hyperbola; genuineness is the separate condition $\Delta_3 \neq 0$. This places the whole family on the classical six-cell classification of Table 3: B_∞ chooses the column (Δ_2) and the point configuration chooses the row (Δ_3).

TABLE 3. The six conic types as infinity gauge (Δ_2) $\times$ point configuration (Δ_3), reproducing the classical classification [6].

	$\Delta_2 > 0$ ($\alpha\beta < 0$)	$\Delta_2 = 0$ ($\alpha\beta = 0$)	$\Delta_2 < 0$ ($\alpha\beta > 0$)
$\Delta_3 \neq 0$	ellipse / circle	parabola	hyperbola
$\Delta_3 = 0$	point	two parallel lines	two crossing lines

Tilted (non-axis-aligned) conics follow by applying a rotation versor (Section 8) to (16), or directly from off-axis ideal directions in B_∞ (general v_∞ factors). Three-dimensional quadric-surface variants are out of scope for this two-dimensional foundation and belong to the QCGA lift discussed in the conclusion.

4.4. Axis-aligned conics by dual construction

Dually, an axis-aligned conic is the grade-1 vector $s = \mathbf{c}^*$ of (13) with $C = 0$. Splitting the origin part with $e_o = e_{o_1} + e_{o_2}$ and $\bar{e}_o = e_{o_1} - e_{o_2}$, so that $-2A e_{o_1} - 2B e_{o_2} = -(A+B) e_o - (A-B) \bar{e}_o$,

$$s = -(A+B) e_o - (A-B) \bar{e}_o + D e_1 + E e_2 - F e_\infty. \quad (18)$$

A point lies on the conic iff $q \cdot s = 0$ (10). For the centred conics ($D = E = 0$, $F = -1$) this gives the closed forms of Table 4; the circle is the familiar centre-point form $p_c - \frac{r^2}{2} e_\infty$.

TABLE 4. Axis-aligned conics by dual construction (grade-1 IPNS vector). The ellipse/hyperbola/parabola are centred at the origin; a centre (c_x, c_y) adds $-2Ac_x e_1 - 2Bc_y e_2$ and sets $F = Ac_x^2 + Bc_y^2 - 1$.

Type	dual vector s
circle (c, r)	$p_c - \frac{r^2}{2} e_\infty$
ellipse (a, b)	$-\left(\frac{1}{a^2} + \frac{1}{b^2}\right)e_o - \left(\frac{1}{a^2} - \frac{1}{b^2}\right)\bar{e}_o + e_\infty$
hyperbola (a, b)	$-\left(\frac{1}{a^2} - \frac{1}{b^2}\right)e_o - \left(\frac{1}{a^2} + \frac{1}{b^2}\right)\bar{e}_o + e_\infty$
parabola $y^2=4px$	$-e_o + \bar{e}_o - 4p e_1$
parabola $x^2=4py$	$-e_o - \bar{e}_o - 4p e_2$

The anisotropic round point. The circle's $p_c - \frac{1}{2}r^2 e_\infty$ generalises by giving the point two independent weights, one per conformal origin. For weights w_1, w_2 define the *anisotropic point* at (c_x, c_y) as one independently weighted bundle per axis,

$$p^{(w_1, w_2)} = w_1(e_{o_1} + c_x e_1 + \frac{c_x^2}{2} e_{\infty_1}) + w_2(e_{o_2} + c_y e_2 + \frac{c_y^2}{2} e_{\infty_2}) + w_1 w_2 c_x c_y e_{\infty_3}, \quad (19)$$

which is *null for every* w_1, w_2 — the per-axis pairings $e_{o_i} \cdot e_{\infty_i} = -1$ cancel axis by axis — and obeys, for a test point $q = q(x, y)$, the weighted distance identity $q \cdot p^{(w_1, w_2)} = -\frac{1}{2}[w_1(x - c_x)^2 + w_2(y - c_y)^2]$. Attaching a radius yields a *single dual vector for every axis-aligned conic*, the *anisotropic round point*

$$\sigma(c_x, c_y, r, w_1, w_2) = w_1(e_{o_1} + c_x e_1 + \frac{c_x^2}{2} e_{\infty_1}) + w_2(e_{o_2} + c_y e_2 + \frac{c_y^2}{2} e_{\infty_2}) + w_1 w_2 c_x c_y e_{\infty_3} - \frac{1}{2}r^2 e_\infty \quad (20)$$

whose locus $q \cdot \sigma = 0$ is the conic $w_1(x - c_x)^2 + w_2(y - c_y)^2 = r^2$, with quadratic coefficients $A = w_1$, $B = w_2$ and square $\sigma^2 = \frac{1}{2}r^2(w_1 + w_2)$. The **weight ratio** $w_1 : w_2 = 1/a^2 : 1/b^2$ **fixes the** (a, b) **shape** and r fixes the size, so (20) realises *every* axis-aligned conic: $w_1 = w_2$ gives a circle, $w_1, w_2 > 0$ an ellipse (semi-axes a, b for $w_i = 1/a_i^2$, $r = 1$), and $w_2 < 0$ a hyperbola. The sign of the radius term is the reality ($-\frac{1}{2}r^2 e_\infty$ the real conic, $+\frac{1}{2}r^2 e_\infty$ its imaginary partner), exactly as for the circle, whose ordinary round point $p_c - \frac{1}{2}r^2 e_\infty$ is the isotropic case $w_1 = w_2 = 1$. Expression (20) agrees with the canonical vector of Table 4 up to the gauge-inert $\bar{e}_\infty, e_{\infty_3}$ terms, which do not move the locus.

Remark 4.2 (why CCGA, and why this is still a conic). CGA's round point $p_c - \frac{1}{2}r^2 e_\infty$ has a single weight and can carry only an isotropic radius (circles); CCGA's *two* conformal origins e_{o_1}, e_{o_2} let the round point carry an *anisotropic* radius, i.e. an ellipse or hyperbola. The weights are exactly the conic's quadratic coefficients $(A, B) = (w_1, w_2)$, and allowing the off-diagonal weight on e_{o_3} adds the cross term C , recovering the general (tilted) dual conic. A “fully weighted point” is therefore a quadratic form — a conic —

rather than a new primitive: the isotropic point of Section 3 remains the base element of the algebra, and σ is read as a *derived* round object.

Proposition 4.3 (type from the origin balance). *In (18) the conic type is set by the balance of the combined and differential origins, $\bar{e}_o/e_o$: writing $A = 1/a^2$ and $B = \pm 1/b^2$,*

$$|A-B| < |A+B| \text{ (ellipse)}, \quad |A-B| = |A+B| \text{ (parabola)}, \quad |A-B| > |A+B| \text{ (hyperbola)}, \quad (21)$$

the circle being the isotropic limit $\bar{e}_o = 0$ ($A = B$), and the parabola the case where one of A, B vanishes.

This is the dual of Proposition 4.1: the wedge construction fixes the type by the *infinity* gauge B_∞ (the conic’s ideal points), whereas the dual construction fixes it by the *origin* blades $e_o, \bar{e}_o$ (its shape) — the gauge duality “shape on the origin side, size on the infinity side” of Section 2.3. The circle’s clean centre-point form $p_c - \frac{r^2}{2}e_\infty$ is exactly the isotropic case where the $\bar{e}_o$ term drops out, while the “ $\pm$ ” that distinguishes ellipse from hyperbola is the sign carried by the $\bar{e}_o$ anisotropy term. Off-centre conics follow by a translator versor (Section 8), and the degenerate members ($\Delta_3 = 0$) are the same special-position limits as on the wedge side.

5. The Object Taxonomy

5.1. The construction ladder

Every CCGA object is a wedge of points with at most the two gauge blades $I_o^\triangleright$ (origin) and $I_\infty^\triangleright$ (infinity, the circular-point pair). Three ladders organise the whole zoo, distinguished by which gauge they carry.

1. **Bare multipole ladder** (no gauge): the plain wedge of k points, $p_1 \wedge \cdots \wedge p_k$, grade k for $k = 1, \dots, 5$ (point, twopole, tripole, quadpole, pentapole). Five points already fix a conic as a point-locus; the pentapole is the last rung.
2. **$I_o^\triangleright$ -gauged conic ladder:** wedging $I_o^\triangleright$ promotes a blade by two grades and gauge-fixes it so the dual is clean (Section 4). For $n \leq 4$ points $p_1 \wedge \cdots \wedge p_n \wedge I_o^\triangleright$ (grade $n+2$) is a *pencil* — the gauge-fixed family of conics through the n points; the $n = 2$ rung is the *gauged dipole* (the conic-gauge point pair, the natural conic $\vee$ line result) and the $n = 4$ rung is the conic $\vee$ conic intersection blade. *A full CCGA conic is precisely a grade-7 blade ending in $\wedge I_o^\triangleright$.* The general conic is $p_1 \wedge \cdots \wedge p_5 \wedge I_o^\triangleright$; the smooth named conics are *three points + an ideal-point pair* $+ I_o^\triangleright$, the pair being two real ideal points (hyperbola, $\Delta_2 > 0$), one double ideal point (parabola, tangent to I_∞ , $\Delta_2 = 0$), or two imaginary ideal points (ellipse, $\Delta_2 < 0$). When that pair is the circular points $I_\infty^\triangleright$ the ellipse becomes a circle — which is therefore a CGA object (next ladder), not listed here.
3. **CGA embedded in CCGA** (the $I_\infty^\triangleright$ sub-array): wedging the circular-point pair $I_\infty^\triangleright$ embeds the standard CGA round/flat family inside CCGA at

grade = (CGA grade) + 2. Round point $p \wedge I_\infty^\flat$, point pair $p_1 \wedge p_2 \wedge I_\infty^\flat$, circle $p_1 \wedge p_2 \wedge p_3 \wedge I_\infty^\flat$, with the flat versions (flat point, line) carrying an extra e_∞ . These are CGA objects, not full CCGA conics.

5.2. Master object table

TABLE 5. The CCGA object zoo, sorted by OPNS grade, in three construction blocks: the bare multipole ladder, the $I_o^\flat$ -gauged conic ladder, and the $I_\infty^\flat$ CGA-embedded sub-array; followed by flats / ideal elements. The ideal-point pair of a smooth conic is written $v_\infty^{(1)}, v_\infty^{(2)}$ (real/imaginary) or v_∞, v'_∞ (double); “CGA gr.” is the grade inside the CGA sub-algebra (CCGA grade - 2).

Object	OPNS form	gr(OPNS)	gr(IPNS)	Reality
<i>Bare multipole ladder (points only)</i>				
Point	p	1	7	null
Twopole	$p_1 \wedge p_2$	2	6	$\text{sign}(\cdot)^2$
Tripole	$p_1 \wedge p_2 \wedge p_3$	3	5	—
Quadpole	$p_1 \wedge \cdots \wedge p_4$	4	4	—
Pentapole	$p_1 \wedge \cdots \wedge p_5$	5	3	—
<i>$I_o^\flat$-gauged conic ladder (a CCGA conic ends in $\wedge I_o^\flat$)</i>				
Pencil ($n \leq 4$)	$p_1 \wedge \cdots \wedge p_n \wedge I_o^\flat$	$n+2$	$6-n$	—
Conic (general)	$p_1 \wedge \cdots \wedge p_5 \wedge I_o^\flat$	7	1	Δ_2, Δ_3
Hyperbola	$p_1 \wedge p_2 \wedge p_3 \wedge v_\infty^{(1)} \wedge v_\infty^{(2)} \wedge I_o^\flat$	7	1	$\Delta_2 > 0$
Parabola	$p_1 \wedge p_2 \wedge p_3 \wedge v_\infty \wedge v'_\infty \wedge I_o^\flat$	7	1	$\Delta_2 = 0$
Ellipse	$p_1 \wedge p_2 \wedge p_3 \wedge v_\infty^{(1)} \wedge v_\infty^{(2)} \wedge I_o^\flat$	7	1	$\Delta_2 < 0$
<i>CGA embedded in CCGA (the $I_\infty^\flat$ sub-array)</i>				
Round point	$p \wedge I_\infty^\flat$	3	—	CGA gr. 1
Point pair	$p_1 \wedge p_2 \wedge I_\infty^\flat$	4	—	CGA gr. 2
Flat point (CGA)	$p \wedge e_\infty \wedge I_\infty^\flat$	4	—	CGA gr. 2
Circle (CGA)	$p_1 \wedge p_2 \wedge p_3 \wedge I_\infty^\flat$	5	—	CGA gr. 3
Line (CGA)	$p_1 \wedge p_2 \wedge e_\infty \wedge I_\infty^\flat$	5	—	CGA gr. 3
<i>Flats / ideal elements</i>				
Ideal point	(drop e_o part)	1	—	n/a
Line at infinity	I_∞	3	—	n/a
Flat point	$p \wedge I_\infty$	4	—	n/a
Conic at infinity	$I_o^\flat \wedge I_\infty$	5	—	n/a
Pseudoscalar	I	8	0	—

The ideal points $v_\infty = v_\infty(v_x, v_y) \in I_\infty$ are the asymptotic directions the conic meets on the line at infinity; the ellipse uses the imaginary pair encoded by the real bivector $(a^2 e_{\infty_1} - b^2 e_{\infty_2}) \wedge e_{\infty_3}$ (a weighted $I_\infty^\flat$, which for $a = b$ degenerates to the circular points and hence to the CGA circle).

5.3. Reality conditions

6. Operations: Join, Meet, and Dual

7. Geometric-Algebra-Native Feature Extraction

8. Transformations as Versors

9. Conic Intersections

9.1. Meet grade arithmetic

9.2. The universal pipeline

9.3. Shared-factor / Bézout reduction

10. Conclusion and Future Work

References

- [1] Breuils, S., Fuchs, L., Hitzer, E., Nozick, V., Sugimoto, A.: Three-dimensional quadrics in extended conformal geometric algebras of higher dimensions from control points, implicit equations and axis alignment. *Advances in Applied Clifford Algebras* **29**(3) (2019). DOI 10.1007/s00006-019-0974-z. URL <https://hal.science/hal-02169419>. Publisher: Springer Verlag
- [2] Chomicki, C., Breuils, S., Biri, V., Nozick, V.: Intersection of conic sections using geometric algebra. In: CGI ENGAGE 2023. Shanghai, China (2023). URL <https://hal.science/hal-04210085>
- [3] Dorst, L., Fontijne, D., Mann, S.: *Geometric Algebra for Computer Science: An Object-Oriented Approach to Geometry*. Elsevier (2010)
- [4] Hrdina, J., Navrat, A., Vašík, P.: *Geometric Algebra for Conics*. *Advances in Applied Clifford Algebras* **28** (2018). DOI 10.1007/s00006-018-0879-2
- [5] Perwass, C.: *Geometric Algebra with Applications in Engineering*, vol. 4. Springer Berlin, Heidelberg (2009). DOI 10.1007/978-3-540-89068-3
- [6] Richter-Gebert, J.: *Perspectives on Projective Geometry: A Guided Tour Through Real and Complex Geometry*. Springer Science & Business Media (2011). Google-Books-ID: F_NP8Kub2XYC

Enzo Harquin and Stéphane Breuils and Venceslas Biri and Vincent Nozick

Enzo Harquin, Venceslas Biri, Vincent Nozick

Université Gustave Eiffel, CNRS, LIGM,

Champs-sur-Marne, F-77420, France

{enzo.harquin2,venceslas.biri,vincent.nozick}@univ-eiffel.fr and

Stéphane Breuils

Université Savoie Mont-Blanc, CNRS, LAMA,

Chambéry, F-73000, France

stephane.breuils@univ-smb.fr